In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.preprocessing import LabelEncoder, MultiLabelBinarizer
from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score
```

In [2]:
```
df = pd.read_csv("movie_metadata (1) (4).csv")  # Load dataset/content/movie_metadata (1) (4).csv
print(df.head())  # View first few rows
```

```
   color      director_name  num_critic_for_reviews  duration  \
0  Color      James Cameron                   723.0     178.0
1  Color     Gore Verbinski                   302.0     169.0
2  Color         Sam Mendes                   602.0     148.0
3  Color  Christopher Nolan                   813.0     164.0
4    NaN        Doug Walker                     NaN       NaN

   director_facebook_likes  actor_3_facebook_likes      actor_2_name  \
0                      0.0                   855.0  Joel David Moore
1                    563.0                  1000.0     Orlando Bloom
2                      0.0                   161.0      Rory Kinnear
3                  22000.0                 23000.0    Christian Bale
4                    131.0                     NaN        Rob Walker

   actor_1_facebook_likes        gross                           genres  ...  \
0                  1000.0  760505847.0  Action|Adventure|Fantasy|Sci-Fi  ...
1                 40000.0  309404152.0         Action|Adventure|Fantasy  ...
2                 11000.0  200074175.0        Action|Adventure|Thriller  ...
3                 27000.0  448130642.0                  Action|Thriller  ...
4                   131.0          NaN                      Documentary  ...

  num_user_for_reviews language  country  content_rating       budget  \
0               3054.0  English      USA           PG-13  237000000.0
1               1238.0  English      USA           PG-13  300000000.0
2                994.0  English       UK           PG-13  245000000.0
3               2701.0  English      USA           PG-13  250000000.0
4                  NaN      NaN      NaN             NaN          NaN

   title_year actor_2_facebook_likes imdb_score  aspect_ratio  \
0      2009.0                  936.0        7.9          1.78
1      2007.0                 5000.0        7.1          2.35
2      2015.0                  393.0        6.8          2.35
3      2012.0                23000.0        8.5          2.35
4         NaN                   12.0        7.1           NaN

  movie_facebook_likes
0                33000
1                    0
2                85000
3               164000
4                    0

[5 rows x 28 columns]
```

In [3]:
```
print(df.isnull().sum())  # Check for missing values
df.dropna(inplace=True)  # Remove null values
```

```
color                       19
director_name              104
num_critic_for_reviews      50
duration                    15
director_facebook_likes    104
actor_3_facebook_likes      23
actor_2_name                13
actor_1_facebook_likes       7
gross                      884
genres                       0
actor_1_name                 7
movie_title                  0
num_voted_users              0
cast_total_facebook_likes    0
actor_3_name                23
facenumber_in_poster        13
plot_keywords              153
movie_imdb_link              0
num_user_for_reviews        21
language                    14
country                      5
content_rating             303
budget                     492
title_year                 108
actor_2_facebook_likes      13
imdb_score                   0
aspect_ratio               329
movie_facebook_likes         0
dtype: int64
```

In [4]:
```
print(df.describe())  # Summary statistics
```

```
       num_critic_for_reviews     duration  director_facebook_likes  \
count             3755.000000  3755.000000              3755.000000
mean                167.392277   110.260186               807.463648
std                 123.465521    22.649332              3068.570417
min                   2.000000    37.000000                 0.000000
25%                  77.000000    96.000000                11.000000
50%                 139.000000   106.000000                64.000000
75%                 224.000000   120.000000               235.000000
max                 813.000000   330.000000             23000.000000

       actor_3_facebook_likes  actor_1_facebook_likes         gross  \
count             3755.000000             3755.000000  3.755000e+03
mean                771.484953             7753.390146  5.262614e+07
std                1894.460316            15520.897163  7.032249e+07
min                   0.000000                0.000000  1.620000e+02
25%                 194.500000              745.000000  8.301882e+06
50%                 436.000000             1000.000000  3.009311e+07
75%                 691.000000            13000.000000  6.690181e+07
max               23000.000000           640000.000000  7.605058e+08

       num_voted_users  cast_total_facebook_likes  facenumber_in_poster  \
count     3.755000e+03                3755.000000           3755.000000
mean      1.058489e+05               11530.158988              1.377630
std       1.520496e+05               19123.805685              2.041689
min       9.100000e+01                   0.000000              0.000000
25%       1.966300e+04                1920.500000              0.000000
50%       5.397700e+04                4060.000000              1.000000
75%       1.286110e+05               16243.000000              2.000000
max       1.689764e+06              656730.000000             43.000000

       num_user_for_reviews        budget   title_year  \
count           3755.000000  3.755000e+03  3755.000000
mean             336.914514  4.624810e+07  2002.974434
std              411.258897  2.260393e+08     9.888558
min                4.000000  2.180000e+02  1927.000000
25%              110.000000  1.000000e+07  1999.000000
50%              210.000000  2.500000e+07  2004.000000
75%              398.500000  5.000000e+07  2010.000000
max             5060.000000  1.221550e+10  2016.000000

       actor_2_facebook_likes   imdb_score  aspect_ratio  movie_facebook_likes
count             3755.000000  3755.000000   3755.000000           3755.000000
mean              2022.314248     6.464740      2.110951           9349.396272
std               4545.393731     1.055865      0.353093          21464.027749
min                  0.000000     1.600000      1.180000              0.000000
25%                385.000000     5.900000      1.850000              0.000000
50%                686.000000     6.600000      2.350000            227.000000
75%                976.000000     7.200000      2.350000          11000.000000
max             137000.000000     9.300000     16.000000         349000.000000
```

In [5]:
```
print("Total variables:", df.shape[1])  # Number of columns
```

```
Total variables: 28
```

In [6]:
```
print(df['language'].value_counts())  # Count of languages
print(df['director_name'].value_counts())  # Count of directors
```

```
language
English       3598
French          34
Spanish         23
Mandarin        15
German          10
Japanese        10
Cantonese        7
Italian          7
Portuguese       5
Hindi            5
Korean           5
Norwegian        4
Danish           3
Thai             3
Persian          3
Dutch            3
Indonesian       2
Dari             2
Aboriginal       2
Aramaic          1
Hungarian        1
Kazakh           1
Maya             1
Filipino         1
Mongolian        1
Czech            1
Russian          1
Zulu             1
Hebrew           1
Arabic           1
Vietnamese       1
Bosnian          1
Romanian         1
Name: count, dtype: int64
director_name
Steven Spielberg    25
Woody Allen         19
Clint Eastwood      19
Ridley Scott        17
Martin Scorsese     16
                    ..
Noel Marshall        1
Todd Lincoln         1
Julian Jarrold       1
Peter Farrelly       1
Shane Carruth        1
Name: count, Length: 1658, dtype: int64
```

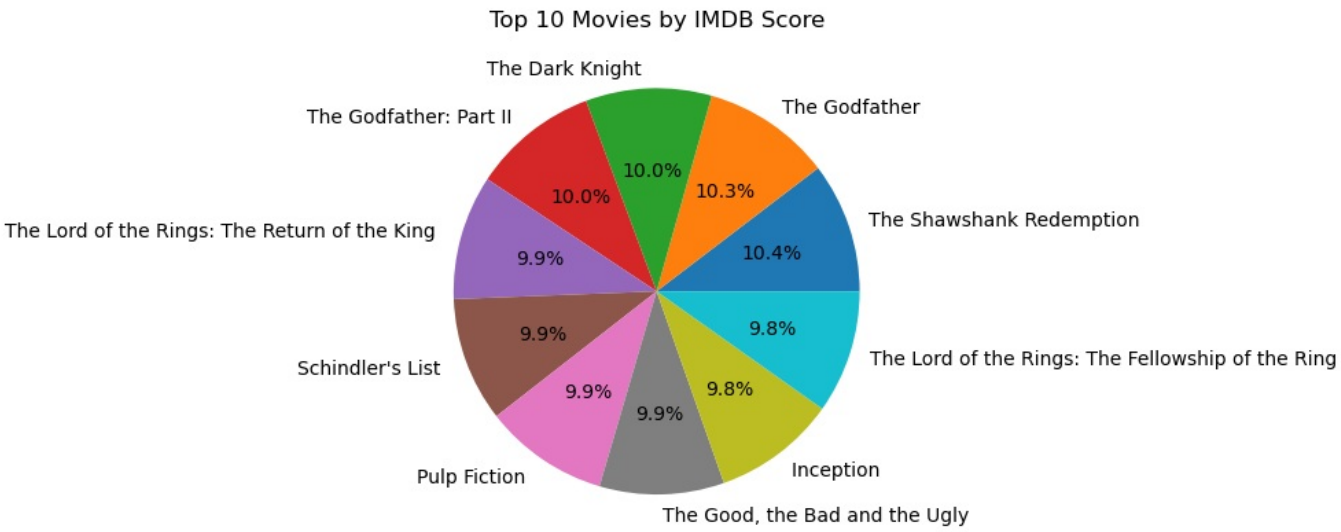
In [7]:
```
hindi_movies = df[df['language'] == 'Hindi']
print(hindi_movies.head())  # Display Hindi movies
hindi_movies.to_csv("hindi_movies.csv", index=False)  # Save for further analysis
```

```
      color director_name  num_critic_for_reviews  duration  \
3075  Color   Karan Johar                    20.0     193.0
3276  Color   Yash Chopra                    50.0     176.0
3344  Color   Karan Johar                   210.0     128.0
4385  Color  Ritesh Batra                   195.0     104.0
4490  Color     Mira Nair                   137.0     114.0

      director_facebook_likes  actor_3_facebook_likes    actor_2_name  \
3075                    160.0                   860.0    John Abraham
3276                    147.0                  1000.0    Katrina Kaif
3344                    160.0                    81.0  Jimmy Shergill
4385                     25.0                    73.0     Nimrat Kaur
4490                    300.0                    73.0   Randeep Hooda

      actor_1_facebook_likes       gross                    genres  ...  \
3075                  8000.0   3275443.0                     Drama  ...
3276                  8000.0   3047539.0             Drama|Romance  ...
3344                  8000.0   4018695.0  Adventure|Drama|Thriller  ...
4385                   638.0   4231500.0             Drama|Romance  ...
4490                   307.0  13876974.0      Comedy|Drama|Romance  ...

     num_user_for_reviews language  country  content_rating       budget  \
3075                264.0    Hindi    India               R  700000000.0
3276                286.0    Hindi    India       Not Rated    7217600.0
3344                235.0    Hindi    India           PG-13   12000000.0
4385                162.0    Hindi    India              PG    1000000.0
4490                214.0    Hindi    India               R    7000000.0

      title_year actor_2_facebook_likes imdb_score  aspect_ratio  \
3075      2006.0                1000.0        6.0          2.35
3276      2012.0                3000.0        6.9          2.35
3344      2010.0                 327.0        8.0          2.35
4385      2013.0                  85.0        7.8          2.35
4490      2001.0                 209.0        7.4          1.85

     movie_facebook_likes
3075                  659
3276                12000
3344                27000
4385                16000
4490                    0

[5 rows x 28 columns]
```

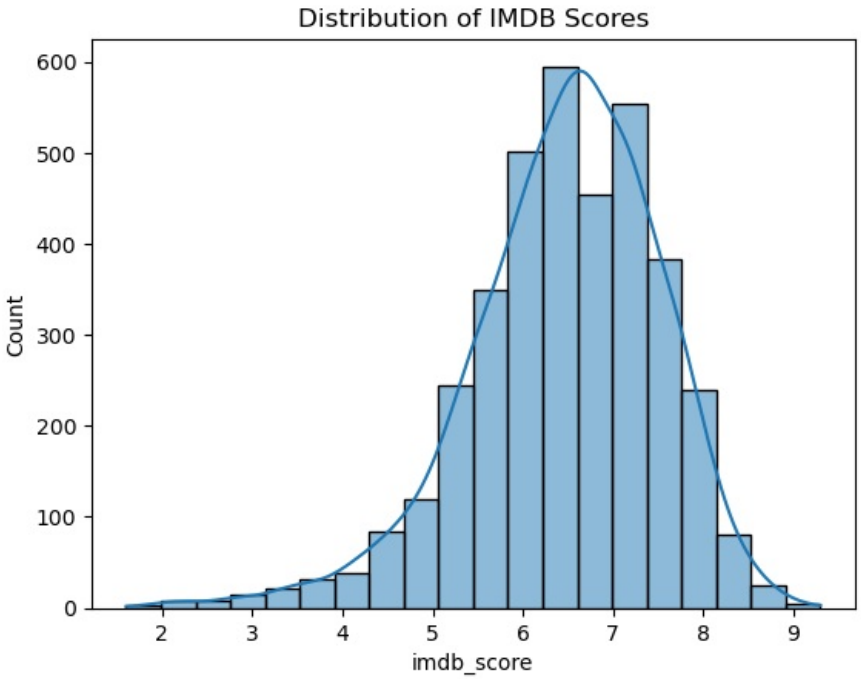
In [8]:
```
top_movies = df[df['imdb_score'] >= 8]
print(top_movies.head())  # Display top-rated movies
```

```
    color      director_name  num_critic_for_reviews  duration  \
3   Color  Christopher Nolan                   813.0     164.0
17  Color        Joss Whedon                   703.0     173.0
27  Color      Anthony Russo                   516.0     147.0
43  Color        Lee Unkrich                   453.0     103.0
47  Color       Bryan Singer                   539.0     149.0

    director_facebook_likes  actor_3_facebook_likes        actor_2_name  \
3                   22000.0                 23000.0      Christian Bale
17                      0.0                 19000.0   Robert Downey Jr.
27                     94.0                 11000.0  Scarlett Johansson
43                    125.0                   721.0   John Ratzenberger
47                      0.0                 20000.0      Peter Dinklage

    actor_1_facebook_likes        gross  \
3                  27000.0  448130642.0
17                 26000.0  623279547.0
27                 21000.0  407197282.0
43                 15000.0  414984497.0
47                 34000.0  233914986.0

                                    genres  ... num_user_for_reviews  \
3                          Action|Thriller  ...                2701.0
17                   Action|Adventure|Sci-Fi  ...                1722.0
27                   Action|Adventure|Sci-Fi  ...                1022.0
43  Adventure|Animation|Comedy|Family|Fantasy  ...                 733.0
47   Action|Adventure|Fantasy|Sci-Fi|Thriller  ...                 752.0

   language  country  content_rating       budget  title_year  \
3   English      USA           PG-13  250000000.0      2012.0
17  English      USA           PG-13  220000000.0      2012.0
27  English      USA           PG-13  250000000.0      2016.0
43  English      USA               G  200000000.0      2010.0
47  English      USA           PG-13  200000000.0      2014.0

   actor_2_facebook_likes imdb_score  aspect_ratio movie_facebook_likes
3                 23000.0        8.5          2.35               164000
17                21000.0        8.1          1.85               123000
27                19000.0        8.2          2.35                72000
43                 1000.0        8.3          1.85                30000
47                22000.0        8.0          2.35                82000

[5 rows x 28 columns]
```

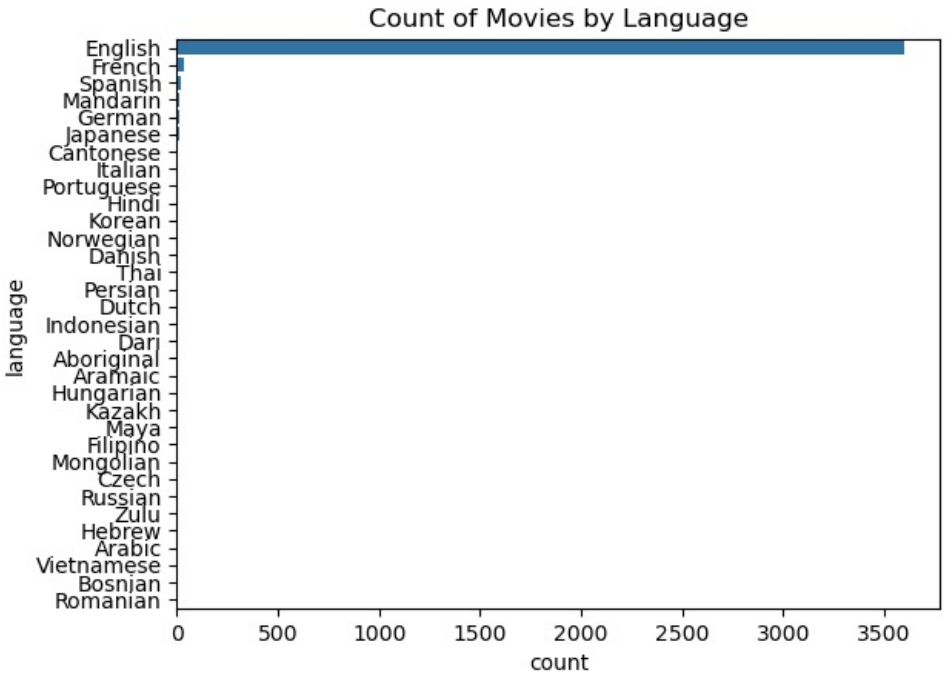
In [9]:
```
top_10 = df.nlargest(10, 'imdb_score')
plt.pie(top_10['imdb_score'], labels=top_10['movie_title'], autopct='%1.1f%%')
plt.title("Top 10 Movies by IMDB Score")
plt.show()
```



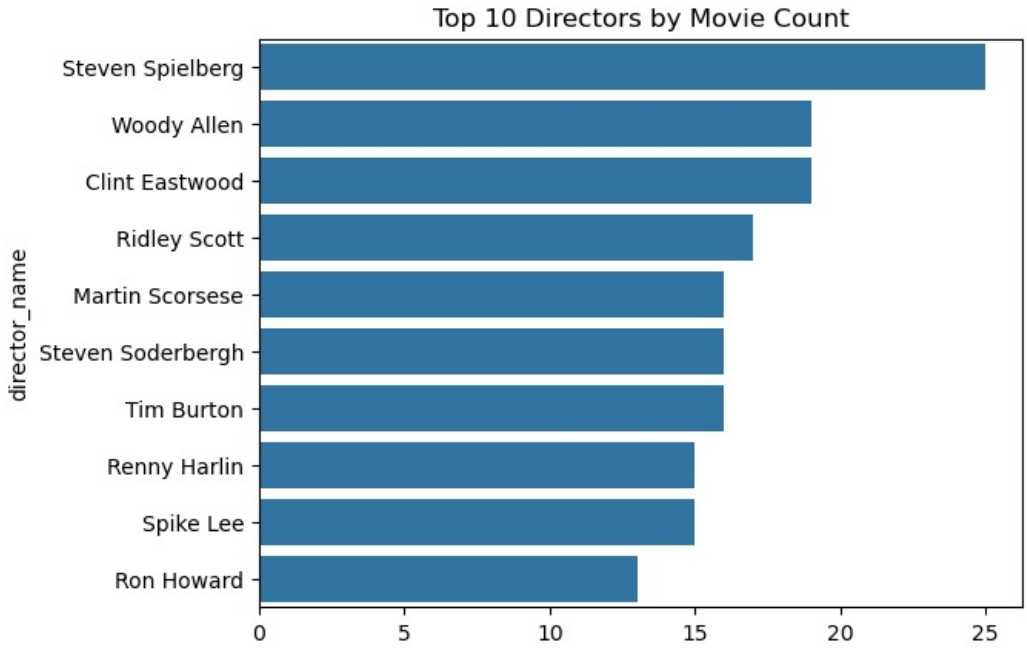
In [10]:
```
sns.histplot(df['imdb_score'], bins=20, kde=True)
plt.title("Distribution of IMDB Scores")
plt.show()
```



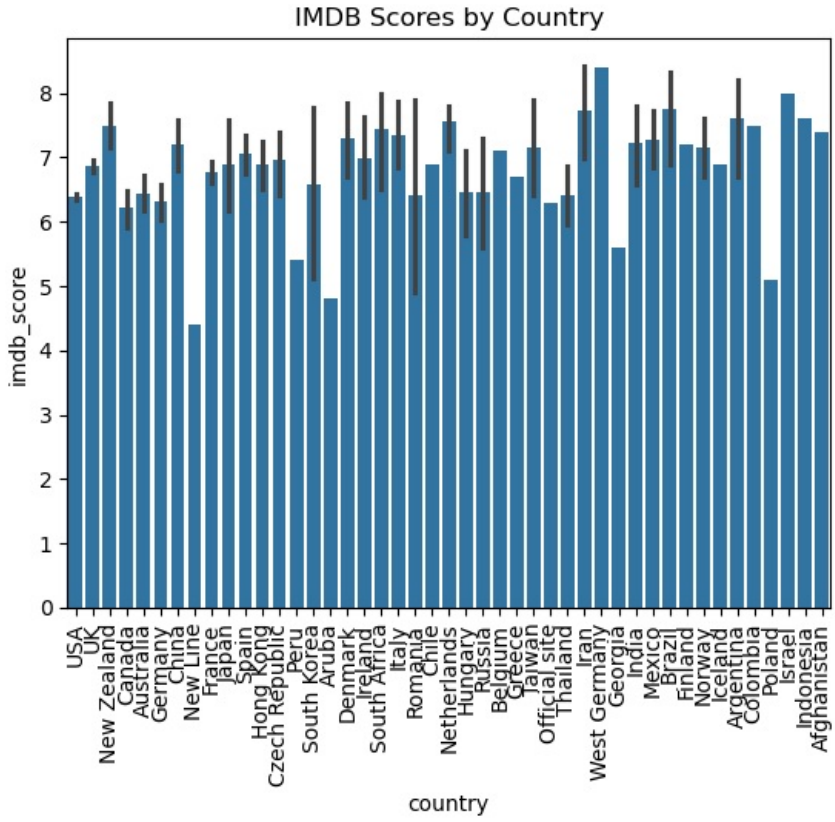
In [11]:
```
sns.countplot(y=df['language'], order=df['language'].value_counts().index)
plt.title("Count of Movies by Language")
plt.show()
```



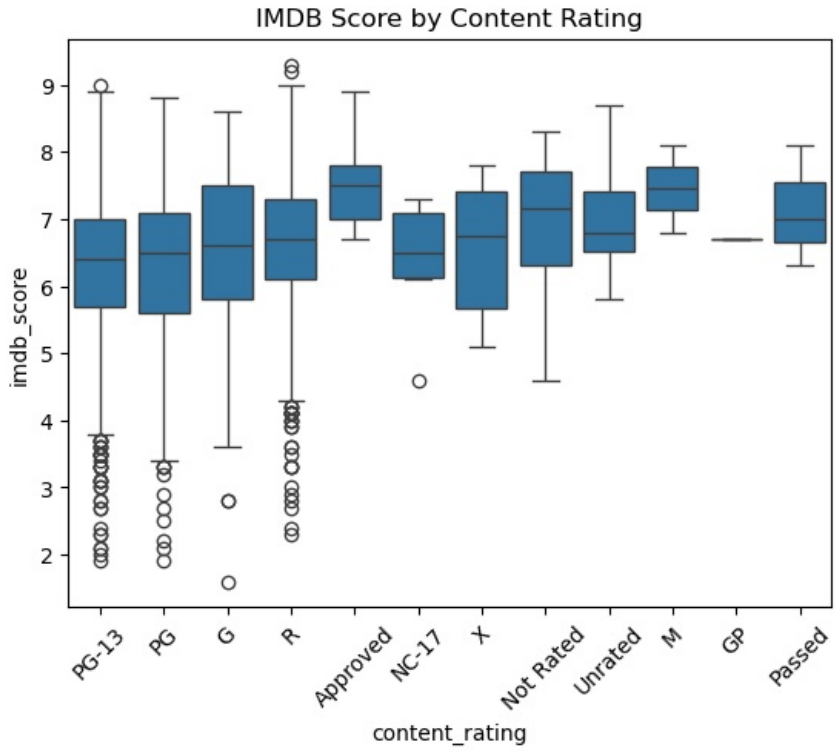
In [12]:
```
top_directors = df['director_name'].value_counts().nlargest(10)
sns.barplot(y=top_directors.index, x=top_directors.values)
plt.title("Top 10 Directors by Movie Count")
plt.show()
```



In [13]:
```
sns.barplot(x=df['country'], y=df['imdb_score'])
plt.xticks(rotation=90)
plt.title("IMDB Scores by Country")
plt.show()
```



In [14]:
```
sns.boxplot(x=df['content_rating'], y=df['imdb_score'])
plt.xticks(rotation=45)
plt.title("IMDB Score by Content Rating")
plt.show()
```



In [15]:
```
 #Check for null values
df.fillna(df.median(numeric_only=True), inplace=True)  # Fill numeric NaNs with median
df.dropna(subset=['director_name', 'language', 'country'], inplace=True)  # Drop rows with missing categorical
```

In [16]:
```
label_encoder = LabelEncoder()
categorical_cols = ['color', 'director_name', 'actor_2_name', 'actor_1_name', 'actor_3_name', 'language', 'coun
for col in categorical_cols:
    df[col] = label_encoder.fit_transform(df[col].astype(str))
```

In [17]:
```
mlb_genres = MultiLabelBinarizer()
genres_encoded = mlb_genres.fit_transform(df['genres'].astype(str).str.split('|'))
genres_df = pd.DataFrame(genres_encoded, columns=mlb_genres.classes_)
df = pd.concat([df, genres_df], axis=1)
df.drop(['genres'], axis=1, inplace=True)

mlb_keywords = MultiLabelBinarizer()
keywords_encoded = mlb_keywords.fit_transform(df['plot_keywords'].astype(str).str.split('|'))
keywords_df = pd.DataFrame(keywords_encoded, columns=mlb_keywords.classes_)
df = pd.concat([df, keywords_df], axis=1)
df.drop(['plot_keywords'], axis=1, inplace=True)
```

In [18]:
```
df.drop(['movie_imdb_link', 'aspect_ratio', 'movie_title'], axis=1, inplace=True)
```

In [19]:
```
X = df.drop('imdb_score', axis=1)
y = df['imdb_score']
```

In [20]:
```
X.fillna(0, inplace=True)
```

In [21]:
```
y.fillna(0, inplace=True)
```

In [22]:
```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

In [23]:
```
models = {
    'Linear Regression': LinearRegression(),
    'Decision Tree': DecisionTreeRegressor(),
    'Random Forest': RandomForestRegressor()
}

for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    print(f"{name} Performance:")
    print("MSE:", mean_squared_error(y_test, y_pred))
    print("MAE:", mean_absolute_error(y_test, y_pred))
    print("R2 Score:", r2_score(y_test, y_pred))
    print("-"*40)
```

```
Linear Regression Performance:
MSE: 1.2294232832178662
MAE: 0.8150588634255107
R2 Score: 0.8192266299255695
----------------------------------------
Decision Tree Performance:
MSE: 0.7800437158469945
MAE: 0.5774863387978142
R2 Score: 0.8853030252119813
----------------------------------------
Random Forest Performance:
MSE: 0.38403586666666656
MAE: 0.41353005464480874
R2 Score: 0.9435316877478679
----------------------------------------
```

In [ ]: